Unit: 6 Unit 2: Exponents Radicals
Topic: Operations with Radicals

Name: ________________
Date: ____________

Foundation (Jalapeno)

Q1. What is $2\sqrt{2} + 3\sqrt{2}$?
(A) $5\sqrt{2}$
(B) $\sqrt{2}$
(C) $4\sqrt{2}$
(D) $6\sqrt{2}$
(E) $7\sqrt{2}$

Q2. Simplify $\sqrt{16} - \sqrt{9}$
(A) 0
(B) 1
(C) 3
(D) 4
(E) -1

Q3. What is $\sqrt{8} \cdot \sqrt{2}$?
(A) 4
(B) 2
(C) $\sqrt{16}$
(D) $\sqrt{4}$
(E) $\sqrt{8}$

Q4. Simplify $(2 + \sqrt{3})(2 - \sqrt{3})$
(A) 1
(B) $4 - 3$
(C) $4 + 3$
(D) $4 - \sqrt{3}$
(E) $2 + 2\sqrt{3}$

Q5. Simplify $6\sqrt{3} + 2\sqrt{3}$
(A) $4\sqrt{3}$
(B) $8\sqrt{3}$
(C) $6\sqrt{2}$
(D) $3\sqrt{2}$
(E) $8\sqrt{2}$

Q6. Simplify $(\sqrt{5})^2$
(A) 25
(B) 5
(C) 10
(D) $\sqrt{25}$
(E) $\sqrt{10}$

Q7. What is $2\sqrt{2} \cdot 3\sqrt{2}$?
(A) 12
(B) $6\sqrt{2}$
(C) $6\sqrt{4}$
(D) $4\sqrt{2}$
(E) $8\sqrt{2}$

Q8. Simplify $\sqrt{12}$
(A) $\sqrt{4} \cdot \sqrt{3}$
(B) $2\sqrt{6}$
(C) $2\sqrt{3}$
(D) $\sqrt{2} \cdot \sqrt{6}$
(E) $\sqrt{2} \cdot \sqrt{3}$

Open Ended Questions (FRQ)

Q9. Simplify $(\sqrt{2} + \sqrt{3})(\sqrt{2} - \sqrt{3})$ and show your work.

Q10. If $a = \sqrt{2}$ and $b = \sqrt{3}$, simplify $a^2 + 2ab + b^2$ and show your work.

Chef's Tips

- Remind students to simplify radicals when possible.
- Encourage students to use the conjugate to rationalize the denominator when necessary.
- Emphasize the importance of showing all work for free response questions.